

SECTION B (30 Marks)

Answer **two (2)** questions from this section.

8. (a) Briefly explain the following concepts as applied in chemical Equilibrium. (02marks)
- Dynamic equilibrium
 - Law of mass action
- (b) For the formation of SO_3 in the gas phase
- $$2\text{SO}_{2(g)} + \text{O}_{2(g)} \rightleftharpoons 2\text{SO}_{3(g)}$$
- $K_p = 40.5 \text{ atm}^{-1}$ at 800K and $\Delta H = -198\text{kJ/mol}$
- At 27°C will K_p be greater than, less than or equal to K_p given?
 - How will the equilibrium be affected if the volume of the vessel containing three gases is reduced, temperature is kept constant.
 - What is the effect on equilibrium position when 1mole of $\text{He}_{(g)}$ is added to a flask containing SO_2 , O_2 and SO_3 at constant temperature (03marks)
- (c) (i) Hydrogen gas used in Haber process is produced by reacting $\text{CO}_{(g)}$ from burning carbon in high temperature steam. The reaction occurs as follows.
- $$\text{CO}_{(g)} + \text{H}_2\text{O}_{(g)} \rightleftharpoons \text{CO}_{2(g)} + \text{H}_{2(g)}$$
- If the reaction vessel contains equimolar mixture of $\text{CO}_{(g)}$ and $\text{H}_2\text{O}_{(g)}$ at 400°C with partial pressure of 3.1 atm and 3.0 atm respectively. Calculate the partial pressure of $\text{H}_2\text{O}_{(g)}$ at 400°C if K_p is 12.4 (05marks)
- (ii) When solid silver is added to a solution containing Ag^+ , Fe^{2+} and Fe^{3+} with concentrations of 0.3M , 0.1M and 0.5M respectively, the following reversible reaction occurs
- $$\text{Ag}_{(aq)}^+ + \text{Fe}_{(aq)}^{2+} \rightarrow \text{Fe}_{(aq)}^{3+} + \text{Ag}_{(s)}$$
- If K_c for this reaction is 6.9 at 298K calculate the equilibrium concentration of each ion in the solution (05marks)
9. (a) Explain the following
- The treatment of alkyl chlorides with aqueous potassium hydroxide results to the formulation of alcohols but in the presence of alcoholic potassium hydroxide alkenes are the major product.
 - Grignard reagents should be prepared under dry conditions.
 - Bromine water reacts readily with hydroxy benzene but has no reaction with benzene.
 - Silver nitrate solution does not precipitate the chloride in $1 - \text{chloropropane}$ but does so in $1 - \text{chloropropane}$ (06marks)
- (b) (i) Benzene can undergo chemical reactions with the following reagents
- Ethylchloride in the presence of AlCl_3 to give A
 - Methyl propene, H^+ to give E
- (ii) A in (i) above undergo a series of reactions as follows:
- It reacts with a mixture of conc H_2SO_4 and conc HNO_3 to give B. B reacts with chlorine in the presence of AlCl_3 to give compound C. C reacts with acidified potassium permanganate to give D.
- (iii) Compound E in (i) above undergo a series of reactions as follows:-
- It reacts with ethylbromide in the presence of AlCl_3 to give F.
 - F React with acidified potassium permanganate to give G.
- Give the structural formula of all aromatic compounds (09marks)
10. (a) (i) After acquiring the knowledge on lowering soil acidity, briefly explain two factors you will consider before purchasing suitable liming materials.
- (ii) With the aid chemical equations (where necessary briefly explain three (3) inorganic cause off soil acidity
- (b) Describe the meaning of cation exchange capacity and give three significances of cation exchange in the soil (04marks)
- (c) A farmer was advised to supply 150kg of nitrogen on the maize farm per hectare. Calculate the mass of the fertilizer with 75% by mass $(\text{NH}_4)_2\text{SO}_4$ which the farmer has to buy in order to meet the nitrogen demand for 10 hectares farm(06marks)

PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
DARES SALAAM REGION
FORM SIX MOCK EXAMINATION
CHEMISTRY 1
Code: F321/
Time 3:00 Hours
Monday 5th February, 2024 P.M

INSTRUCTIONS

- This paper consists of section **A** and **B** with a total of ten (10) questions
- Answer **all** question in section A and two (2) questions from section B.
- Each question carries **ten (10)** marks in section A and **fifteen (15)** marks in section B.
- Non – programmable calculator may be used.
- Write you **Index Number** on every page of your answer booklet (s)
- For calculator the following constants may be useful
 - Plank's constant, $h = 6.623 \times 10^{-34} \text{ Js}$
 - Velocity of light $c = 3 \times 10^8 \text{ m/s}$
 - Universal gas constant $R = 0.821 \text{ atm L/mol K}$ or 8.314 J/mol K
 - Atomic masses
 $\text{H} = 1, \text{C} = 12, \text{O} = 16, \text{Mg} = 24, \text{Ca} = 40, \text{Na} = 23, \text{Cl} = 35.5, \text{Br} = 80, \text{K} = 39$

GEPAM science hub
WhatsApp 0748978398

This paper consists of 4 printed pages

SECTION A (70 Marks)

Answer **all** question from this section

- Metal hydroxides can be prepared in the laboratory by both direct and indirect method depending on the nature of the hydroxide. Below are the hydroxides which can be prepared by one of the two methods. For each of the given metal hydroxide, state whether it can be prepared by direct or indirect method and give the corresponding balanced chemical equation (s) for each.
 - Sodium hydroxide
 - Magnesium hydroxide
 - Barium hydroxide
 - Iron (ii) hydroxide

(06marks)
 - With the help of chemical equation, state the suitable method for the preparation of the following sulphates.
 - Zinc sulphate
 - Calcium sulphate

(03marks)
 - Outline two (2) uses of metal hydroxides **(01mark)**
- A volatile organic compound weighs. 0.2g. On heating in the Victor Meyer's tube, it displaces 30ml of air at 27°C and 756mmHg. Determine the molecular mass of the compound (aqueous tension at 27°C is equal to 26 mmHg) **(03marks)**
 - In 4 minutes, 16.2cm³ of water vapour effuses through a small hole in the time 8.1cm³ of mixture of NO₂ and N₂O₄ effuse through the same hole. Calculate the percentage by volume of NO₂ in the mixture **(04marks)**
 - Two flasks A and B, have equal volume. Flask A contains H₂ and is maintained at 200k while flask B contains an equal mass of CH₄ gas at 400K. In which flask will the molecules move faster? **(03marks)**
- Comment on the following facts.
 - 1 litre of NaCl_(aq) solution has greater osmotic pressure than equimolar 1 litre of aqueous glucose.
 - Boiling point of a solvent increase by the presence of solute in it.
 - A driver adds ethylene glycol to water in a car radiator during winter season

(03marks)
 - The degree of dissociation of Ca (NO₃)₂ in a dilute aqueous solution containing 5.0g of the salt per 100g water at 100°C is 60%. If the vapour pressure of pure water at 100°C is 760mmHg, calculate the vapour pressure of the solution. **(04marks)**
 - Derive the relationship between molar mass of non – volatile solute and relative lowering of vapour pressure when the solute dissolved in volatile solvent. **(03marks)**
- The electron in an atom is located at a definite distance from the nucleus in an orbit which has discrete energy"
 - Which atomic model in the statement based on?
 - Challenge the above statement according to the wave mechanical model of the atom

(02marks)
 - Give an account for the experiment which led Rutherford to conclude that every atom has heavy positively charged nucleus which occupies a small volume.
 - Calculate the energy emitted when electrons in 1g of hydrogen atoms undergo transitions giving line of lowest energy in visible region of its atomic spectrum
 - Give two example to show that electronic configuration of half - filled and full – filled orbitals are more stable. **(06marks)**

- Differentiate between the following terms
 - Enthalpy of solution and enthalpy of combustion.
 - Enthalpy of sublimation and enthalpy of fusion

(03marks)
 - What will be the rise in temperature of solution when 400cm³ of 0.2M HCl_(aq) is mixed with 600cm³ of 0.2 NaOH. Given that

$$\text{H}^+_{(aq)} + \text{OH}^-_{(aq)} \rightarrow \text{H}_2\text{O} \quad \Delta H = -57.1 \text{ kJ/mol}$$
 And specific heat capacity (c) of water = 4.18Jg⁻¹ K⁻¹ **(03marks)**
 - Given the reaction $\text{C}_3\text{H}_{8(g)} + 5\text{O}_{2(g)} \rightarrow 3\text{CO}_{2(g)} + 4\text{H}_2\text{O}_{(g)}$
 Use the following average bond enthalpies to estimate the enthalpy (ΔH) of combustion of propane.

C – H =	414. kJ/mol
C – C =	347 kJ/mol
O = O =	498 kJ/mol
C = O =	741 kJ/mol
H – O =	646 kJ/mol

(04marks)
- Briefly explain the following terms, giving one example in each case.
 - Hydrogen bond
 - Coordinate bond
 - Polar covalent bond

(03marks)
 - Compare and contrast the following concepts
 - Metallic bond and ionic bond
 - Van der Waals forces and hydrogen bond.

(04marks)
 - Identify the compound with higher melting point in each case of the following pairs. Give reason(s)
 - CH₃CH₂OH and CH₃CHO
 - NaCl and AlCl₃

(03marks)
- The following names are considered to be incorrect according to IUPAC system. Give the correct structural formula and the correct names as per IUPAC system.
 - Methyl butane
 - 2 – isopropyl propane
 - 4 – methylpent – 1 – ene
 - 1 – ethyl – 2,2 – dimethyl propane
 - 2 – propyl but – 2 – ene.

(05marks)
 - Identify by giving structural formula of organic compound A and B

A $\xrightarrow[200^\circ\text{C}]{\text{H}_2, \text{Ni}}$ $\begin{array}{c} \text{CH}_3 \text{CH} \text{CH}_3 \\ | \\ \text{CH}_3 \end{array}$

(i) **(02marks)**

(ii) $\text{B} + \text{H}_2\text{O} \xrightarrow{\text{H}_2\text{SO}_4, \text{HgSO}_4} \text{CH}_3\text{COCH}_3$ **(02Marks)**
 - 10cm³ of a gaseous hydrocarbon x needs 45cm³ of oxygen for complete combustion. X reacts with 1 mole of bromine to give a compound of relative molecular mass 200 which contains 79.2% bromine.
 - Give the structural formula of x
 - Determine the molecular mass of X

(03marks)

PRESIDENT'S OFFICE A.K.
REGIONAL ADMINISTRATION AND LOCAL
GOVERNMENT

FORM SIX MOCK EXAMINATION
2024

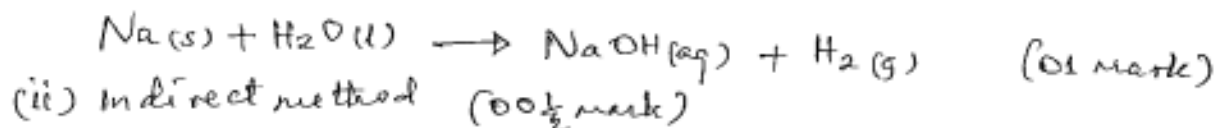
CHEMISTRY PAPER - 1

MARKING SCHEME

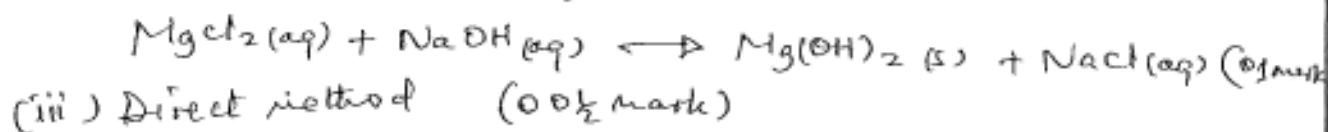
GEPAM science hub

WhatsApp 0748978398

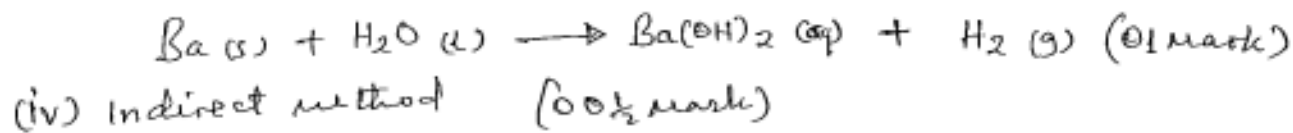
1(a) (i) Direct method (00½ mark)



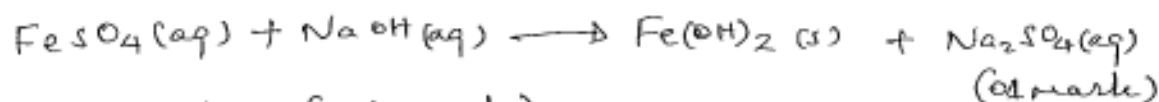
(ii) Indirect method (00½ mark)



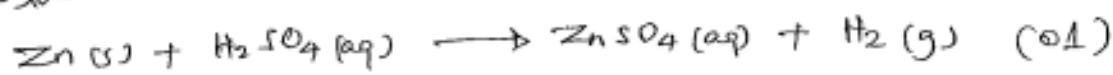
(iii) Direct method (00½ mark)



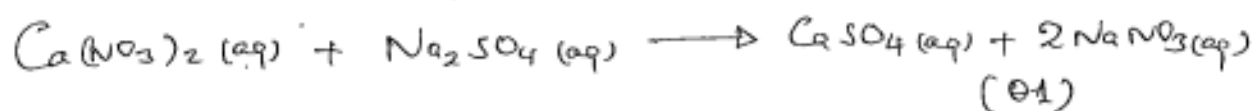
(iv) Indirect method (00½ mark)



(b) Direct method (00½ mark)



(c) Indirect method : (00½)



(c) Uses of metal hydroxides

- Manufacture of Soap eg NaOH and KOH
- As lining materials eg Mg(OH)_2 and Ca(OH)_2
- Extraction of metals eg NaOH
- Softening of hard water eg Ca(OH)_2
- Manufacture of bleaching agents eg NaOH and Ca(OH)_2 .

Any two @ 00½ → (01)

$$2.(a) \text{ Vapour pressure of organic compound} = P_{\text{Total}} - P_{\text{H}_2\text{O}}(\text{atm})$$

$$= 756 - 26$$

$$= 730 \text{ mmHg.} \quad (\text{001})$$

$$PV = \frac{m}{M} RT, \quad M = \frac{mRT}{PV} \quad (\text{001})$$

$$M = \frac{0.23 \times 0.082 \text{ atm dm}^3 \text{ mol}^{-1} \text{ K}^{-1} \times 300 \text{ K}}{730/760 \text{ atm} \times 30 \times 10^{-3} \text{ dm}^3} \quad (\text{001})$$

$$M = 170.7 \text{ g mol}^{-1} \quad (\text{001})$$

$$(b) \frac{V_{\text{water vapour}}}{t} / \frac{V_{\text{mixture}}}{t} = \sqrt{\frac{M_{\text{mixture}}}{M_{\text{water}}}} \quad (\text{001})$$

$$\frac{V_{\text{water vapour}}}{V_{\text{mixture}}} = \sqrt{\frac{M_{\text{mixture}}}{M_{\text{water}}}} \quad (\text{001})$$

$$M_{\text{mixture}} = \left(\frac{V_{\text{water vapour}}}{V_{\text{mixture}}} \right)^2 \times M_{\text{water}} \quad (\text{001})$$

$$= \left(\frac{16.2}{8.1} \right)^2 \times 18 = 72 \text{ g mol}^{-1} \quad (\text{001})$$

$$\text{Let } x \text{ be \% by volume of } \text{NO}_2: \% \text{ of } \text{N}_2\text{O}_4 = (1-x) \times 100\% \quad (\text{001})$$

$$46x + (1-x)92 = 72 \quad (\text{001})$$

$$x = \frac{20}{46} = 0.435 \quad (43.5\%) \quad (\text{001})$$

$$(c) \text{ The average speed } (\bar{u}) \text{ of gas molecules} \propto \sqrt{T/M} \quad (\text{001})$$

$$\bar{u}_A \propto \sqrt{\frac{T_A}{M_A}}, \quad \bar{u}_B \propto \sqrt{\frac{T_B}{M_B}} \quad (\text{001})$$

$$\frac{\bar{u}_A}{\bar{u}_B} = \sqrt{\frac{T_A \cdot M_B}{T_B \cdot M_A}} = \sqrt{\frac{200 \times 16}{400 \times 2}} = 2 \quad (\text{001})$$

$$\bar{u}_A = 2\bar{u}_B; \text{ Molecules in flask A moves with double speed} \quad (\text{001})$$

3(a)(i) NaCl(aq) has greater osmotic pressure than aqueous glucose; NaCl dissociates leading to larger number of particles than in glucose solution.

(ii) Some solvent molecules are prevented by solute particles from escaping into vapour phase. This lowers vapour pressure above the solution.

(iii) Ethylene glycol is the antifreezing agent, it lowers freezing point of water preventing it from freezing as environmental temperature declines.

@ 001 → (003)

(b) $\frac{\Delta p}{p} = i X_{\text{solute}}$ (00½)

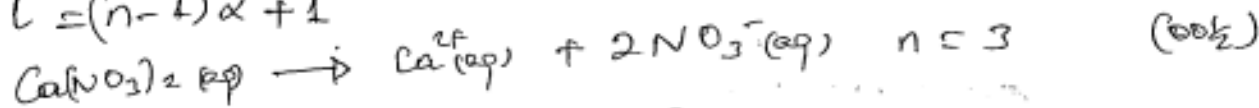
$\frac{\Delta p}{p}$ = relative lowering in vapour pressure (00½)

i = Van't Hoff's factor

$X_{\text{solute}} = \frac{n_{\text{solute}}}{n_{\text{solvent}}}$ since $n_{\text{solute}} \ll n_{\text{solvent}}$ - (00½)

$= \frac{m_{\text{solute}} \times M_{\text{solvent}}}{M_{\text{solute}} \times m_{\text{solvent}}}$ (00½)

$i = (n-1)\alpha + 1$ (00½)



$i = (3-1) \times 0.6 + 1 = 2.2$

$\frac{\Delta p}{760} = \frac{2.2 \times 5 \times 18}{164 \times 100} = 0.012$ (00½)

$\Delta p = 760 \times 0.012 = 9.18 \text{ mmHg}$

Vapour pressure of solution = $p - \Delta p = 760 - 9.18 = 750.82 \text{ mmHg}$ (00½)

(c) $\frac{\Delta p}{p} = \frac{m_{\text{solute}} \times M_{\text{solvent}}}{M_{\text{solute}} \times m_{\text{solvent}}}$ (001)

$M_{\text{solute}} = \frac{m_{\text{solute}} \times M_{\text{solvent}}}{\frac{\Delta p}{p} \times m_{\text{solvent}}}$ (01)

If other parameters are constant: Molar mass of solute is inversely proportional to relative lowering of vapour pressure. (001)

4(a) (i) Bohr's atomic model (00½)

(ii) The model views an electron as being located at certain distance from the nucleus (00½)

It is against the Heisenberg's uncertainty principle which states that "it is impossible to determine precisely both position and velocity of a subatomic particle in motion" (01)

(b) In the Rutherford's α -scattering experiment:

- Most of the α -particles passed through the gold foil without being deflected. (00½)

- Very few α -particles were either deflected to some angle or reflected back. (00½)

This led to the following conclusion:

→ Since some α -particles were deflected, the nucleus is positively charged (00½)

→ Since some α -particles were reflected, the nucleus is heavy

→ Since most of α -particles passed through the Gold foil without being deflected or reflected, the nucleus occupies a small volume of an atom. (00½)

(c) $\Delta E = nhf$

$$\Delta E = nh \frac{c}{\lambda}$$

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

(00½)

$\Delta E = nhc R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$, $n_1 = 2$, $n_2 = 3$ for line of lower energy in visible spectrum.

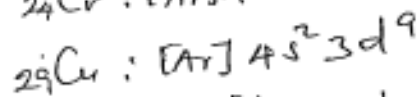
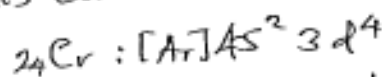
1g of H-atoms $\equiv$ 1 mole of H-atoms equal to 6.02×10^{23} atoms (00½)

$$\Delta E = 6.02 \times 10^{23} \times 6.62 \times 10^{-34} \times 3 \times 10^8 \times 1.097 \times 10^7 \left(\frac{1}{2^2} - \frac{1}{3^2} \right)$$

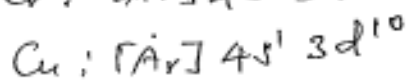
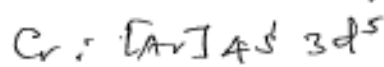
$$= 182345 \text{ mol} = 182.324 \text{ kJ/mol}$$

(00½)

(c) (ii) Consider: Cr and Cu.



Due to stability of half filled and completely filled orbitals, their electronic configurations are:



5(a)(i) Enthalpy of solution is the change in enthalpy when one mole of a solute dissolves in specified amount of solvent at particular temperature. WHILE Enthalpy of combustion is the change in enthalpy when one mole of substance is completely burnt in air (001)

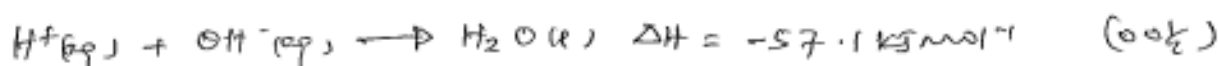
(ii) Enthalpy of sublimation is the change in enthalpy when one mole of solid substance is converted to gas without melting

WHILE Enthalpy of fusion is the change in enthalpy when one mole of a solid substance is converted into liquid state at its melting point. (001)

$$(b) n_{HCl} = \text{Molarity} \times \text{Volume (dm}^3) = 0.2 \text{ mol/L} \times 0.4 \text{ L} = 0.08 \text{ mol}$$

$$n_{NaOH} = 0.2 \times 0.6 = 1.2 \text{ mol} \quad (001)$$

$$n_{H^+} = n_{HCl} = 0.08 \text{ mol}, \quad n_{OH^-} = n_{NaOH} = 1.2 \text{ mol} \quad (001)$$

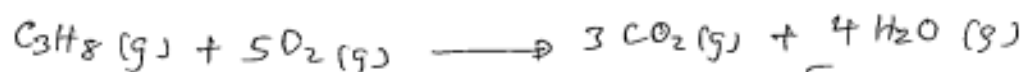


$$H = n_{H_2O} \times \Delta H = 0.08 \text{ mol} \times -57.1 \text{ kJ mol}^{-1} = -4568 \text{ J} \quad (00\frac{1}{2})$$

$$H = -m \times \Delta T \times C \quad (00\frac{1}{2})$$

$$\Delta T = \frac{H}{-mC} = \frac{-4568 \text{ J}}{-1000 \text{ g} \times 4.18 \text{ J g}^{-1} \text{ } ^\circ\text{C}^{-1}} = 1.09 ^\circ\text{C} \quad (00\frac{1}{2})$$

$$5(c) \Delta_r H = \sum \text{bond broken} - \sum \text{bond formed} \quad (001)$$



$$\Delta_r H = [2(C-C) + 8(C-H) + 5(O=O)] - [6(C=O) + 8(O-H)] \quad (001)$$

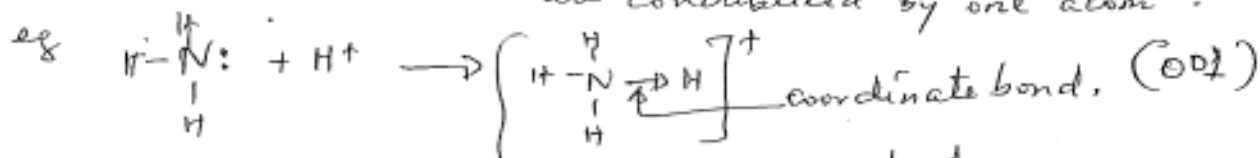
$$\Delta_r H = [2 \times 347 + 8 \times 414 + 5 \times 498] - [6 \times 741 + 8 \times 464] \quad (00\frac{1}{2})$$

$$\Delta_r H = -1662 \text{ kJ mol}^{-1} \quad (00\frac{1}{2})$$

6(a)(i) Hydrogen bond is the dipole-dipole attractive force between hydrogen atom bonded to a highly electronegative atoms such as N, O and F of the same or nearly molecule. (001)



(ii) Coordinate bond: is the type of covalent bond which is formed when all shared electrons are contributed by one atom.



(iii) polar covalent bond: is a type of covalent bond formed between atoms of different electronegativity, thus resulting to the formation of partial charges. eg $\text{H}^{\delta+}-\text{Cl}^{\delta-}$ (001)

(b)(i) Similarity:

• Both metallic bond and ionic bond involves very strong electrostatic attraction holding ions together in crystals. (001)

• Difference

In metallic bond; electrostatic attraction is between metallic positively ions and delocalized electrons of metal atoms WHILE in ionic bond electrostatic attractive force is between oppositely charged ions (001)

(ii) Similarity

→ Both are weak bonding forces

• Difference:

→ Hydrogen bond is stronger than Van der Waals forces

→ Hydrogen bond occur in molecules with hydrogen atoms bonded to N, O or F only. WHILE Van der Waals forces is more broad - being responsible for all covalent compounds whether containing H atoms or not. (001)

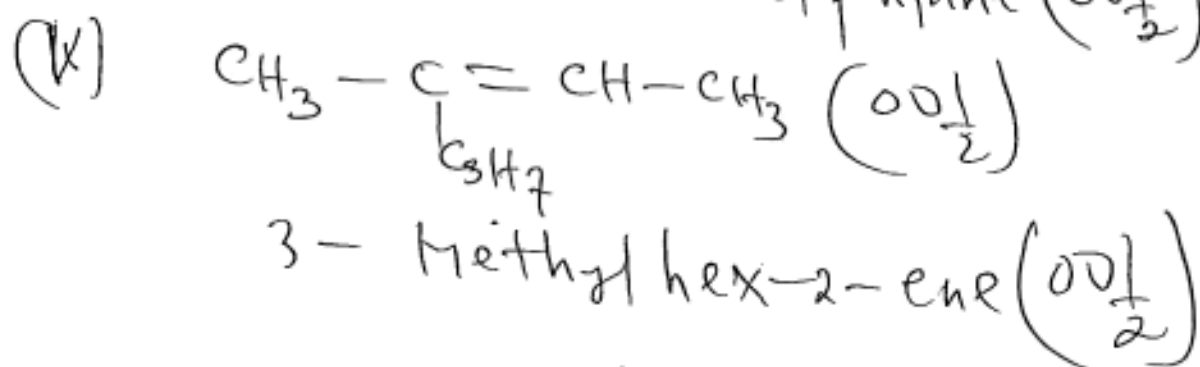
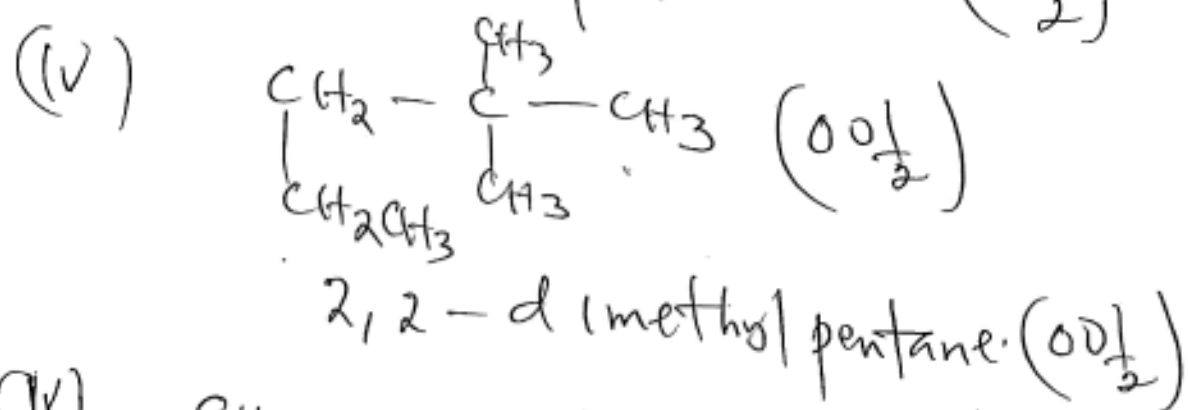
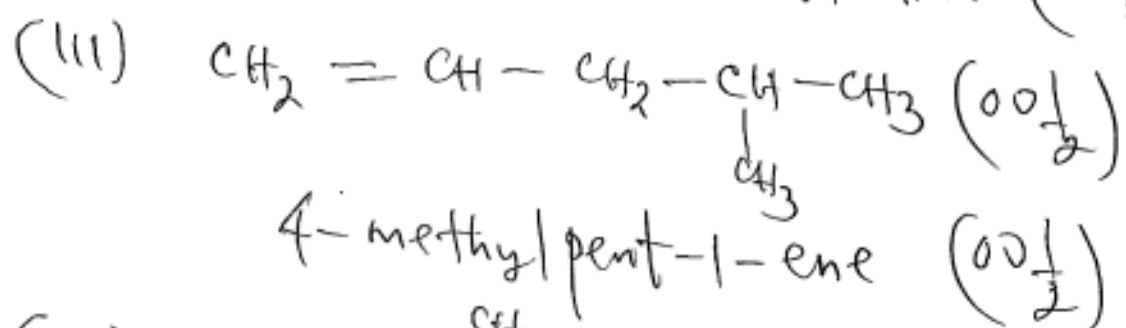
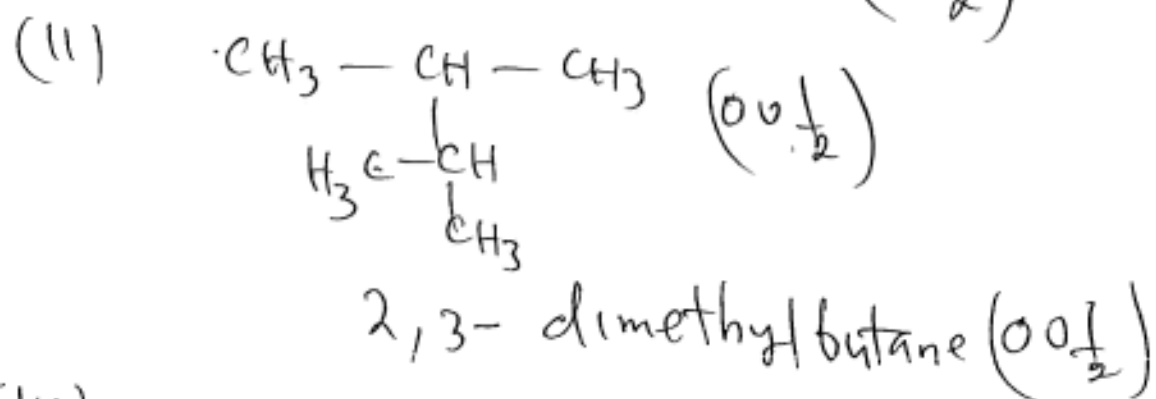
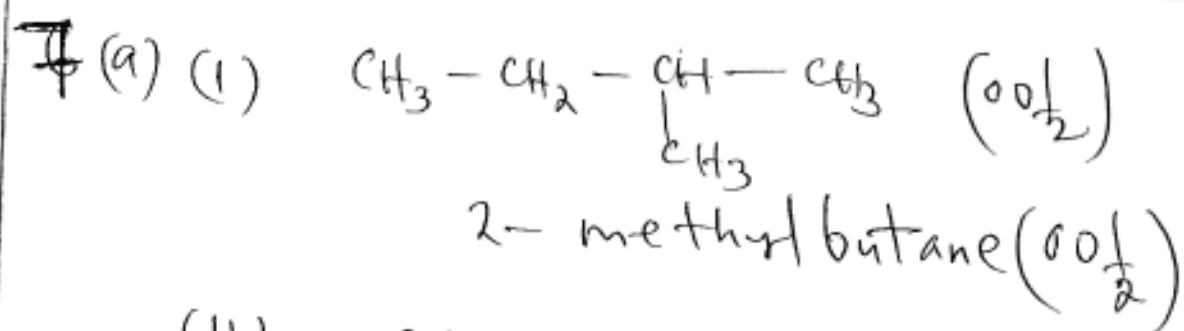
(c) (i) $\text{CH}_3\text{CH}_2\text{OH}$ (001)

Reason

it forms hydrogen bond which increases the strength of the bonds hence higher melting point WHILE CH_3CHO do not form hydrogen bond.

(ii) NaCl (001)

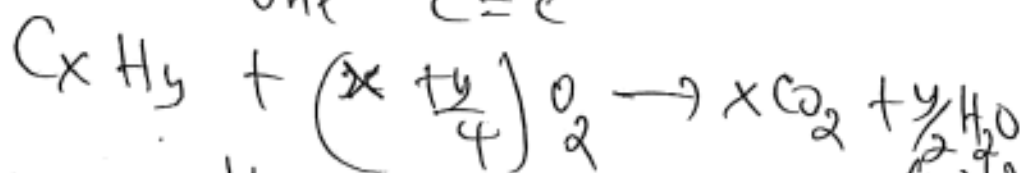
Reason: it is more ionic in character than AlCl_3 due to its small polarizing power of large sized Na^+ than Al^{3+} . (001)



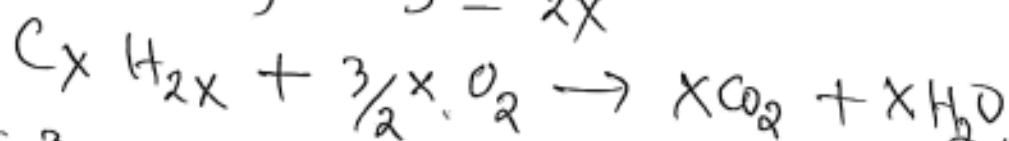
(b)(i) structural formula of A
is $\text{CH}_3 - \underset{\text{CH}_3}{\text{C}} = \text{CH}_2$ (01)

(ii) structural formula of B is
 $\text{CH}_3\text{C}\equiv\text{CH}$ (01)

(c)(i) Since only one mole of Br_2 was added with one $\text{C}=\text{C}$ must be alkene.



For an alkene with only one $\text{C}=\text{C}$, $y = 2x$



10 cm³ of X hydrocarbon (C_xH_{2x}) reacted with 45 cm³ of oxygen (01)

According to Avogadro's law,
Mole ratio = Volume ratio

$$45\text{ cm}^3 = \frac{3}{2}x \times 10$$

$$x = 3, \quad 2x = y$$

$$(2 \times 3) = y$$

$$y = 6$$

$\therefore$ structural formula of X
 C_3H_6 (01)

(II) Molecular Mass
(~~C~~ H₆)

$$(12 \times 3) + (1 \times 6) = 42 \text{ g/mol} \quad \left(\begin{array}{l} 00 \\ 2 \end{array} \right)$$

Alternatively:

— Since X react with only one mole of Br₂ X must be an alkene.

$$\therefore C_n H_{2n} Br_2 = 200 \quad \left(\begin{array}{l} 00 \\ 2 \end{array} \right)$$

Molecular Mass of bromine

$$\frac{79.2}{100} \times 200 = 158.4 \quad \left(\begin{array}{l} 00 \\ 2 \end{array} \right)$$

$$C_n H_{2n} = 200 - 158.4 \quad \left(\begin{array}{l} 00 \\ 2 \end{array} \right)$$

$$12n + 2n = 41.6$$

$$14n = 41.6$$

$$n = \frac{41.6}{14} \approx 3 \quad \left(\begin{array}{l} 00 \\ 2 \end{array} \right)$$

(I) $\therefore$ structural formula of X
C₃H₆ $\left(\begin{array}{l} 00 \\ 2 \end{array} \right)$

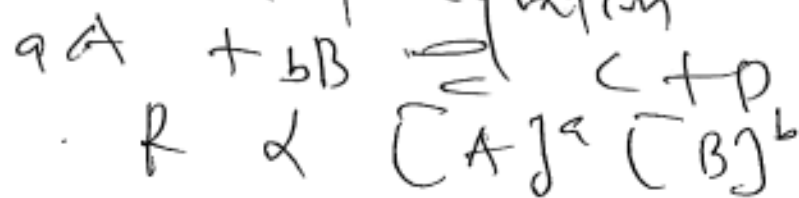
(II) Molecular Mass of X

$$C_3H_6 = (12 \times 3) + (6 \times 1)$$
$$= 42 \text{ g/mol} \quad \left(\begin{array}{l} 00 \\ 2 \end{array} \right)$$

(09)

8(a)(i) Dynamic equilibrium is a state of equilibrium of a reversible rxn at which rate of forward rxn is equal to the rate of backward reaction (01)

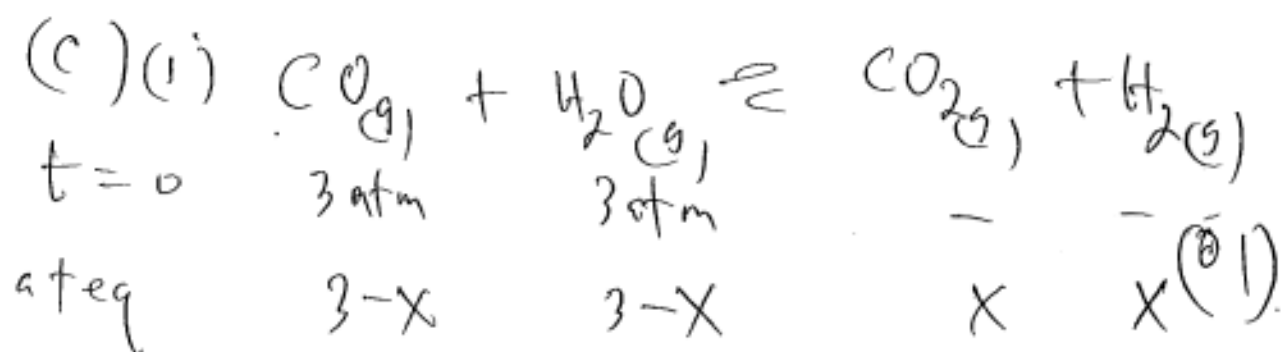
(ii) Law of Mass action states that. At a constant temperature and pressure the rate of chemical rxn is directly proportional to the product of molar concentrations of the reactants each raised to the power equal to corresponding stoichiometric coefficient which appear in the balanced chemical equation (01)



(b)(i) Since rxn is exothermic, its equilibrium constant will increase with lowering in temperature. Hence, value of K_p at 300K is large than K_p at 800K. (01)

(ii) When the volume of the vessel is reduced, the pressure of the mixture will increase. The equilibrium will shift to side of low volume (forward rxn is favoured) (01)

8(b) (iii) Addition of helium (inert gas) at constant temperature and volume has no effect in the equilibrium (01)



$$K_p = \frac{(P'_{\text{CO}_2}) (P'_{\text{H}_2})}{(P_{\text{CO}}) (P_{\text{H}_2\text{O}})} \left(\frac{0.5}{2} \right)$$

$$K_p = \frac{x^2}{(3-x)(3-x)} \left(\frac{0.1}{2} \right)$$

$$12.4 = \frac{x^2}{(3-x)^2}$$

$$x^2 - 6x + 9 = 12.4 \left(\frac{0.5}{2} \right)$$

$$11.4x^2 - 74.4x + 111.6 = 0$$

Solving quadratically.

$$x = 2.3 \text{ atm. (01)}$$

$$\therefore P'_{\text{H}_2\text{O}} = 3 - x$$

$$= 3 - 2.3 = 0.7 \text{ atm.}$$

$$\therefore \text{partial pressure of H}_2\text{O} = 0.7 \text{ atm (01)}$$

8(c)(ii) Reaction quotient Q is given by.

$$Q = \frac{[Fe^{3+}]}{[Ag^+] [Fe^{2+}]^{(0.5/2)}}$$

$$Q = \frac{0.5 \text{ mol/L}}{(0.3 \text{ mol/L}) \times (0.1 \text{ mol/L})}$$

$$Q = \frac{0.5 \text{ mol/L}}{16.7 \text{ mol/L}^{(0.5/2)}} (0.1)$$

$Q > K_c$ rxn proceed to the Left.



$t=0$	0.5	0.3	0.1 $(0.5/2)$
$t=t$	$-x$	$+x$	$+x$
$t \text{ equilibrium}$	$0.5-x$	$0.3+x$	$0.1+x$

$$K_c^{-1} = \frac{1}{K_c} = \frac{[Ag^+] [Fe^{2+}]}{[Fe^{3+}]}$$

$$\frac{1}{6.9} = \frac{(0.3+x)(0.3+x)(0.5/2)}{0.5-x}$$

Solving x quadratically to get

$$x^2 + 0.54x - 0.04 = 0$$

$$x = 0.07 \text{ M} \quad (0.1)$$

$$\begin{aligned}
 \therefore [Al^{3+}] &= 0.3 + x = 0.367M & \left(\begin{smallmatrix} 0.01 \\ 2 \end{smallmatrix} \right) \\
 [Fe^{2+}] &= 0.1 + x = 0.167M & \left(\begin{smallmatrix} 0.01 \\ 2 \end{smallmatrix} \right) \\
 [Fe^{3+}] &= 0.5 - x = 0.433M & \left(\begin{smallmatrix} 0.01 \\ 2 \end{smallmatrix} \right)
 \end{aligned}$$

09(a)(i) Water (H_2O) in aqueous medium being more polar than alcohol, causes more polarization of positively charged carbon atom bonded to chlorine. As the carbon becomes more positively charged, nucleophilic substitution rxn which leads to alcohol formation is more favoured than elimination rxn which would lead to the formation of alkene. (1½)

(ii) In the presence of water (in the solution) the grignard reagent would react to form alkane. So, to prevent the formation of alkane, it must be prepared under dry or anhydrous conditions. (1½)

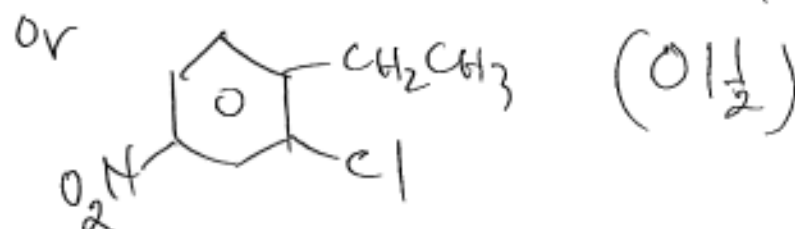
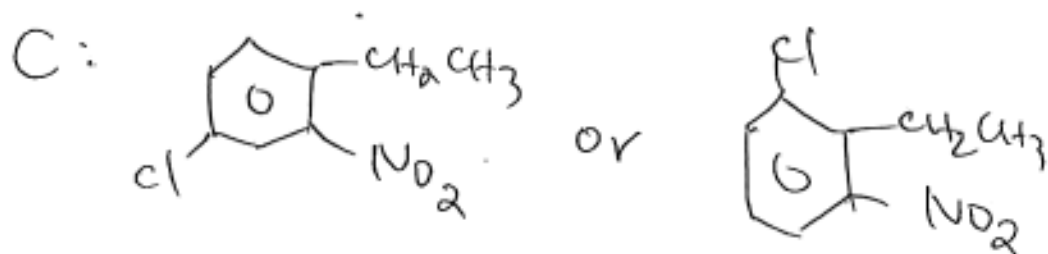
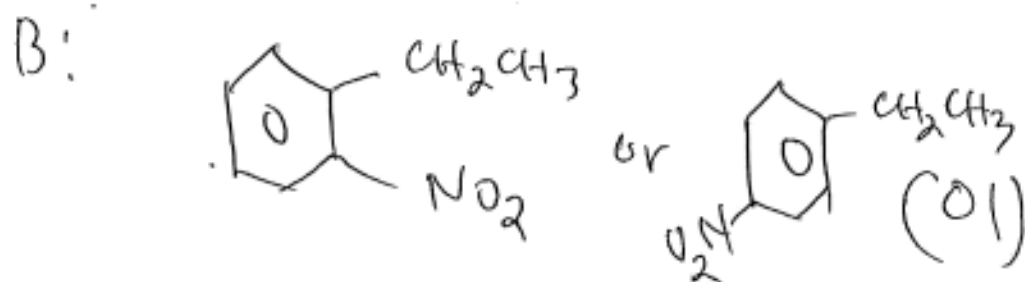
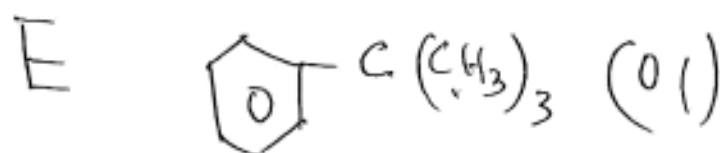
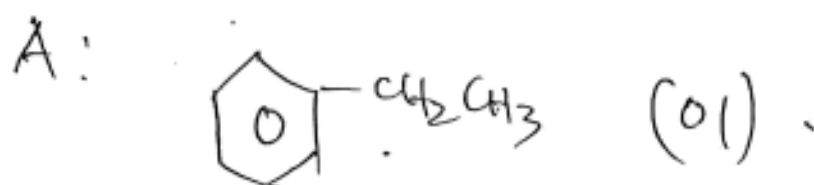
$$R-MgX + H_2O \rightarrow R-H + Mg(OH)X$$

(iii) In hydroxy benzene, electrophilic substitution rxn is activated by presence of OH group which is an activator and thus making it possible to react with bromine water unlike unsubstituted benzene. (1½)

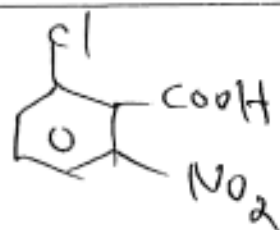
9(a)(iv) To precipitate the chloride, the Cl atom in a given chlorohydrocarbon must be substituted by a nucleophile.

In 1-chloropropane the Cl to be substituted is bonded to a π bonded carbon atom which is nucleophile too. and hence there is very large repulsion b/w the two nucleophiles which prevents the rxn to occur unlike in 1-chloropropane. (1½)

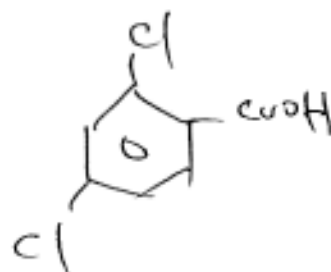
(b)



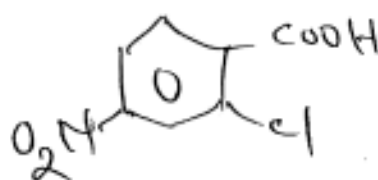
09 (b) D:



or

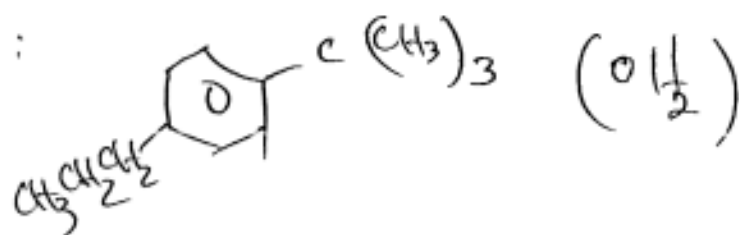


or

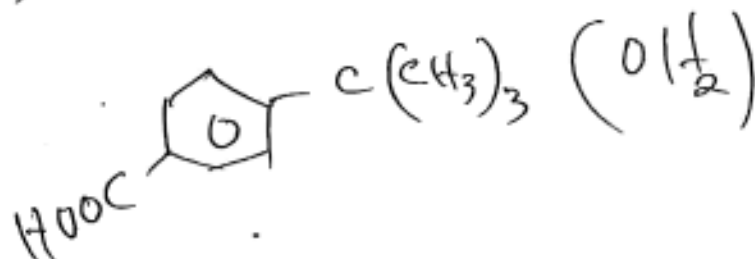


$(0 \frac{1}{2})$

F:



G:



- 10 (a) (i) Amount of Impurity present in the Liming Material. (001/2)
- Liming material with high percentage purity have high neutralizing value. (001/2)
 - Molar Mass of the liming material (001/2)
 - Increase in molar mass decrease neutralization value (001/2)

- (ii) Heavy rains or over-irrigation leach basic metals eg Na, K, Mg etc leaving excess H⁺ in the top of soil. (01)
- Acidic fertilizer eg (NH₄)₂SO₄, HCl which upon hydrolysis add acidity in the soil. (001/2)

$$\text{NH}_4^+ + \text{H}_2\text{O} \rightarrow \text{NH}_3 + \text{H}_3\text{O}^+$$
 - Industrial emission of acidic gases eg CO₂, SO₂, NO₂ etc which add acidity in the soil. (001/2)

eg.

$$\text{SO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{SO}_3 \rightleftharpoons 2\text{H}^+ + \text{SO}_3^{2-}$$

$$\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{CO}_3 \rightleftharpoons 2\text{H}^+ + \text{CO}_3^{2-}$$